

# NEXORA E-Commerce Project Report

## Market-ready storefront and backend architecture overview

NEXORA is a full-stack e-commerce platform built with Next.js, FastAPI, PostgreSQL, and Redis. The application supports customer shopping, wishlist and cart workflows, checkout, admin operations, and live order updates.

The storefront uses a dark luxury visual identity and includes product discovery, quick-view modals, cart drawer interactions, and order/account pages. The backend manages customers, products, orders, delivery snapshots, and access controls with JWT authentication.

The data model centers on Users, Products, Cart/Wishlist records, delivery Addresses, and Order snapshots that preserve historical order data.

## Project Stack

Frontend: Next.js 16 + React + TypeScript + Tailwind CSS + Framer Motion + Three.js

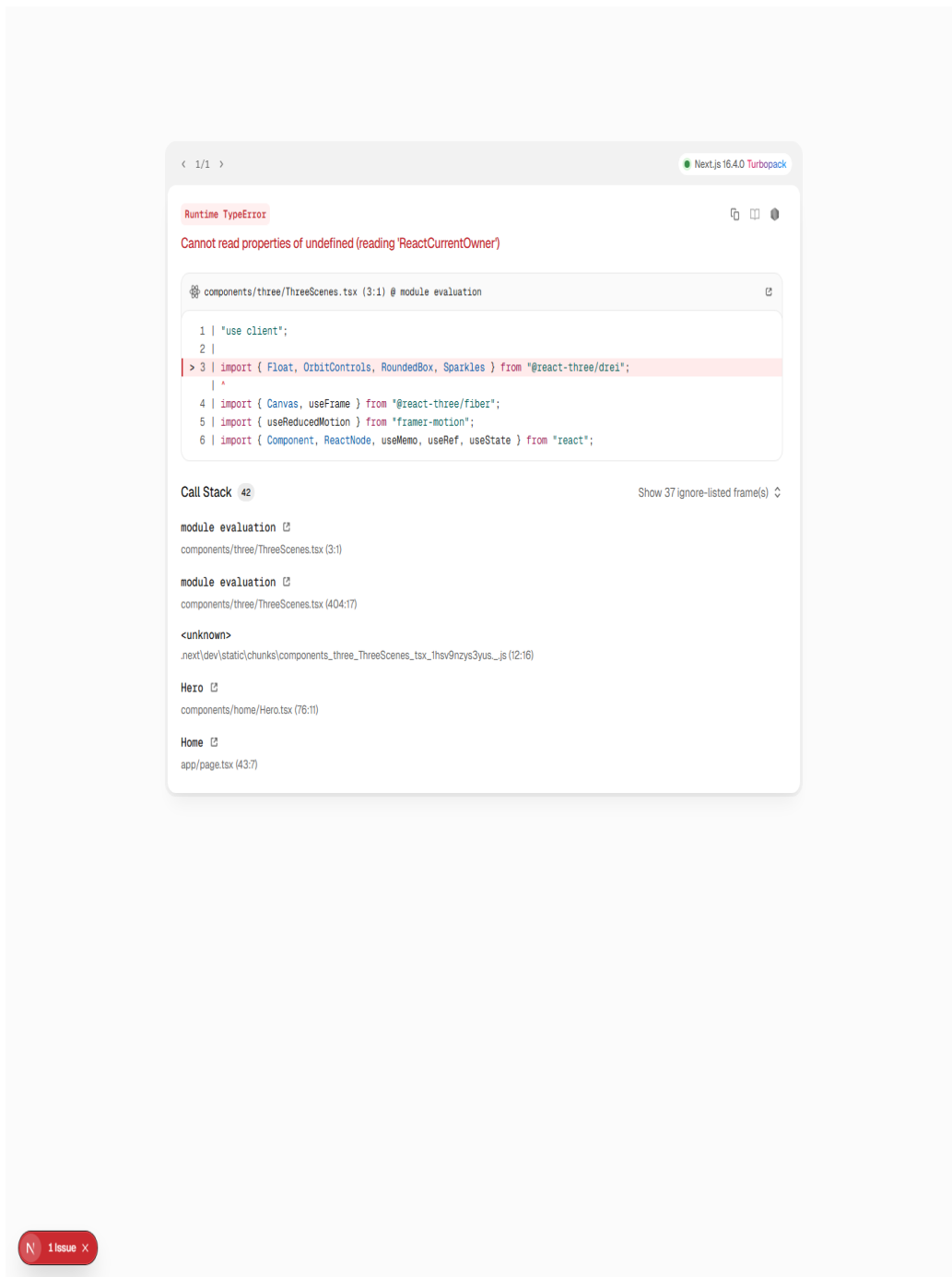
Backend: FastAPI + SQLAlchemy + PostgreSQL + JWT auth + Redis

Operations: Docker support, product catalog APIs, order management, admin dashboard, and WebSocket notifications

## Key Features

- ? Customer authentication with bcrypt-hashed passwords and JWT access tokens
- ? Product browsing with search, filtering, sorting, and category-based product detail pages
- ? Cart and wishlist management persisted for logged-in users
- ? Address management and order snapshots so shipping data remains immutable after checkout
- ? Admin access for product and order operations, with sales/statistics dashboards
- ? Redis caching for product listing and rate limiting for auth endpoints
- ? Real-time order status updates over WebSocket notifications

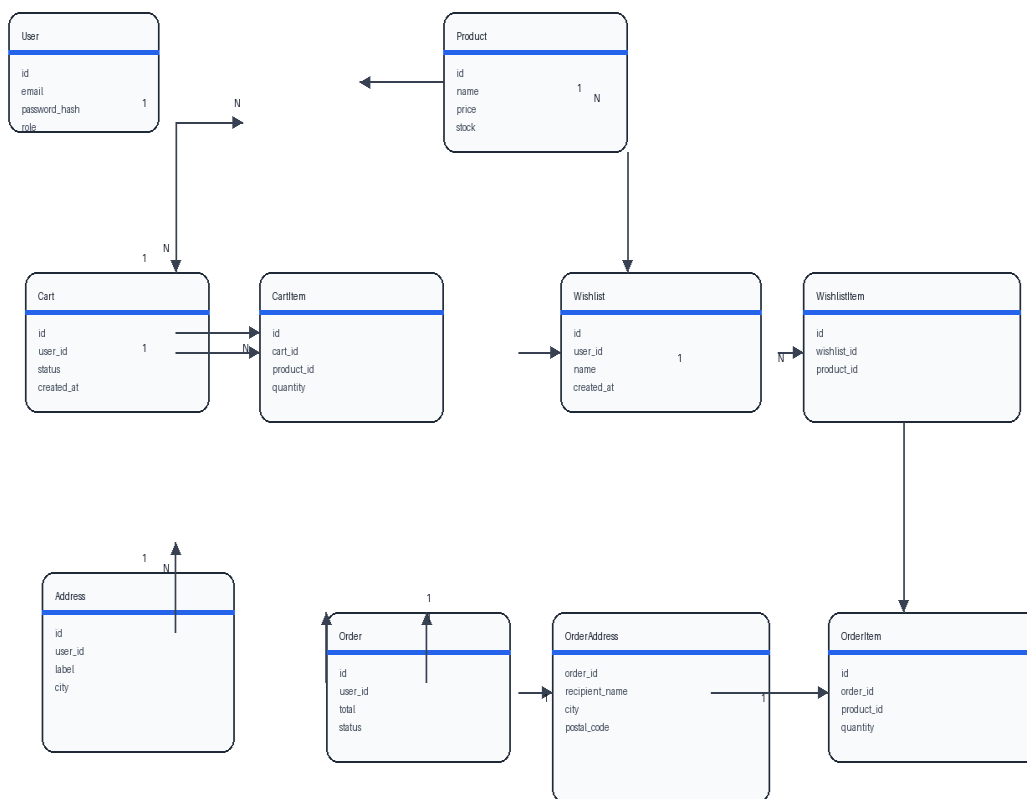
## Captured Storefront Screenshot



## Current storefront status

The landing page currently renders a runtime error in the three.js hero scene: Cannot read properties of undefined (reading "ReactCurrentOwner"). This is associated with a React/Next compatibility issue between the project dependencies and the @react-three/fiber stack.

# Entity Relationship Diagram (ERD)



## ERD Notes

- ? Users own carts, addresses, wishlists, and orders.
- ? Products are linked to cart items, wishlist items, and order items.
- ? Orders store a snapshot of the delivery address in order\_addresses so order records are immutable.
- ? Cart and wishlist records enforce uniqueness to keep product entries consistent.